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Table 1.2: Different Classes of viruses	
Class	**Example**
Class 1 - Viruses with dsDNA	Adenoviruses
Class 2 –Viruses with (+) sense ssDNA	Parvo viruses
Class 3 –Viruses with dsRNA	Reo viruses
Class 4 –Viruses with (+)sense ssRNA	Toga viruses
Class 5 – Viruses with (−)sense ssRNA	Rhabdo viruses
Class 6 – Viruses with (+) sense ss RNA –RT: that replicate with DNA intermediate in life cycle	Retro viruses
Class 7 – Viruses with ds DNA –RT: that replicate with RNA intermediate in life cycle	Hepadna viruses

or double stranded - ds), RNA or DNA, the use of reverse transcriptase(RT), ss RNA may be (+) sense or (−) antisense. Viruses are classified into seven classes (Table 1.2).

Viral genome

Each virus possesses only one type of nucleic acid either DNA or RNA. The nucleic acid may be in a linear or circular form. Generally nucleic acid is present as a single unit but in wound tumour virus and in influenza virus it is found in segments. The viruses possessing DNA are called **'Deoxyviruses'** whereas those possessing RNA are called **'Riboviruses'**. Majority of animal and bacterial viruses are DNA viruses (HIV is the animal virus which possess RNA). Plant viruses generally contain RNA (Cauliflower Mosaic virus possess DNA). The nucleic acids may be single stranded or double stranded. On the basis of nature of nucleic acid viruses are classified into four Categories. They are Viruses with ssDNA (Parvoviruses), dsDNA (Bacteriophages), ssRNA (TMV) and dsRNA(wound tumour virus).

1.2.5 Tobacco Mosaic Virus (TMV)

Tobacco mosaic virus was discovered in 1892 by Dimitry Ivanowsky from the Tobacco plant. Viruses infect healthy plants through vectors like aphids, locusts etc. The first visible symptom of TMV is discoloration of leaf colour along the veins and show typical yellow and green mottling which is the mosaic symptom. The downward curling and distortion of young apical leaves occurs, plant becomes stunted and yield is affected.

Structure

Electron microscopic studies have revealed that TMV is a rod shaped (Figure 1.4b) helical virus measuring about 280x150µm with a molecular weight of $39x10^6$ Daltons. The virion is made up of two constituents, a protein coat called **capsid** and a core called **nucleic acid**. The protein coat is made up of approximately 2130 identical protein subunits called **capsomeres** which are present around a central single stranded RNA molecule. The genetic information necessary for the formation of a complete TMV particle is contained in its RNA. The RNA consists of 6,500 nucleotides.